The quorum exponents:

what reduces, what the classical bounds miss, what remains

D. Joubert

3 September 2026

Abstract

The every-scale quorum asks for two bounds on the bottom vector v of Q_L: $\delta_p = v^\top T_p v \le e^{-7p+c}$ and $\kappa_p = \|T_p v - (v^\top T_p v)v\| \ge e^{-4p-c'}$, uniformly in the window. Both quantities are pairings of v against the prime tower T_p. The leading term is the window autocorrelation of v at lag $\log p$. Bandlimited decay (Slepian, Bernstein, integration by parts) gives powers of $1/\log p$, not e^{-7p}. The measured silence is the decay of a near-kernel, not of a desert prolate. The hole shrinks to one estimate: $\|T_p v\|$ itself is exponentially small in p. That estimate is not proved. Nothing here is RH.

1 Reduction

T_p is the contribution of the powers $p^k \le \mu$ to the prime-side form,

$$T_p(f,f) = \sum_{p^k \le \mu} (\log p)\, p^{-k/2}\, \Theta_f(k \log p), \qquad \Theta_f(y) = \int f(x) f(x+y)\, dx,$$

with Θ_f supported in $[0, L]$, $L = \log \mu$, and $\Theta_f(L) = 0$. Then $\delta_p = T_p(v,v)$ and $\kappa_p^2 = \|T_p v\|^2 - \delta_p^2$. The $k = 1$ term dominates for $p \ge 5$:

$$\delta_p = \frac{\log p}{\sqrt{p}}\, \Theta_v(\log p) + O\big(p^{-1} \Theta_v(2 \log p)\big). \tag{1}$$

So $\delta_p \le e^{-7p+c}$ is a pointwise bound on the autocorrelation of one explicit vector at the points $\log p$.

2 What the classical bounds give

If v were merely unit-norm in PW_τ,

$$|\Theta_v(y)| \le \min\big(1,\, C_N (\tau y)^{-N}\big)$$

after N integrations by parts on a smooth cutoff. At $y = \log p$ this is a power of $1/\log p$. The Slepian that lives in the desert $(-\gamma_1, \gamma_1)$ has $\Theta(y) \sim \gamma_1 \operatorname{sinc}(\gamma_1 y/\pi)$, which at $p = 7$ is $O(1/28)$, against a measured $\delta_7 \sim 10^{-16}$. Duffin–Schaeffer on gaps loses the same exponential that *sampling-floor* already recorded for c_L.

None of these tools sees e^{-7p}. The exponent linear in p is not the Fourier decay of a function bandlimited to γ_1.

3 What the measurements say instead

At $\mu = 11$, $-\ln\delta_p \approx 7p - 12$ for $p \geq 5$, and later $-\ln\delta_p \approx 0.19\, s(\chi)\, p\log p$ across ζ, χ_3, χ_4. Both δ_p and κ_p are exponentially small, so $\|T_p v\|$ itself is exponentially small: v is close to the kernel of T_p, not just orthogonal to $T_p v$ in one direction. That is stronger than silence. It means the near-kernel of Q is already almost orthogonal to every individual tower of a large prime.

This matches the leakage picture: the mass of $\hat{v}$ is expelled to the band edge, and the pairing against a lag $\log p$ — a frequency modulation of width 1 — sees that edge through a rapidly oscillating integral. Writing a bound that tracks the measured $7p$ requires a pointwise description of $\hat{v}$ on the edge, uniformly in L. The leakage profile $\ln|\hat{v}(\gamma)| \approx -\tau(\omega_{\max} - \gamma)$ is measured, not proved, and τ itself moves with μ.

4 An attempt

Θ_v is supported in $[0, L]$. The points $y = \log p$ lie *inside* the support. Exponential decay as $y \to \infty$ is the wrong picture: there is no infinity. What is asked is a dip of size e^{-7p} at each arithmetic lag $\log p \in (0, L)$.

Near the edge. Bernstein plus $\Theta_v(L) = 0$ gives $|\Theta_v(\log p)| \leq \tau\log(\mu/p)$. Useful only for $p = \mu^{1-o(1)}$. At $p = 7$, $\mu = 11$, the bound is $O(1)$ against 10^{-16}.

Contour shift. $F \in PW_\tau$ is entire of type τ, so $\int F(\omega)\overline{F(\omega)}\, e^{i\omega y}\, d\omega$ can be shifted. The shift produces a factor $e^{-\delta y}$ and a bound $e^{\tau\delta}$ on the new line — net $e^{(-\delta+\tau)y}$ at best, i.e. a power of p, and only after replacing $|F|^2$ on $\mathbb{R}$ by $F(z)F(\bar{z})$ with control we do not have on the ground state. Super-exponential in $p = e^y$ never comes from a strip.

Fixed p, $L \to \infty$. The measured δ_p is stable from $\mu = 8$ to 16. The statement uniform in μ is therefore: the Beurling extremal v_L of the zero set in $PW_{L/2}$ has $\Theta_{v_L}(\log p)$ bounded by e^{-7p} for each fixed p, for all large L. That is a quantitative property of one sampling-extremal function attached to Γ. It is the same order of difficulty as a quantitative lower bound for c_L, already recorded as out of reach of Duffin–Schaeffer / Bernstein (*sampling-floor*, §4).

5 Two lemmas that do hold

Lemma 1 (edge)**.** $\Theta_v(L) = 0$*, hence the tower of a prime sitting at the window edge vanishes identically (*$T_{11} \equiv 0$ *at* $\mu = 11$*).*

Lemma 2 (crude)**.** $|\delta_p| \leq (\log p)\, p^{-1/2}\|v\|_1^2$*. In particular* $\delta_p = O(p^{-1/2}\log p)$*.*

Lemma 2 is weaker than the 2×2 criterion by an exponential factor, exactly as Bernstein was weaker than c_L by $e^{L\gamma_1/2}$.

6 Why $p\log p$, why a Gaussian

The linear laws $-ln\delta \sim 7p$ and $-ln\kappa \sim 4p$ were local slopes. In the weight $w = p\log p$ the measurements collapse (§87):

$$-\ln\delta_p \approx 0.19\, s(\chi)\, w, \qquad -\ln\kappa_p \approx c(\mu)\, w^2 \quad (p \geq 5).$$

One constant 0.19 for ζ, χ_3, χ_4 at four windows. $c(\mu)$ grows with the window (0.005–0.03). The 2×2 margin $\ln(\kappa^2/\delta) \approx 0.19 s\, w - 2c\, w^2$ is then a downward parabola: it peaks near $w \approx 43$

and would vanish near $w \approx 86$ ($p \approx 27$ at χ_3, $\mu = 30$), which is the measured finite range of the single-vector certificate.

Why $w = p \log p$. $y = \log p$ is the additive lag at which Θ_v is evaluated. $e^y = p$ is the multiplicative place of the same prime. The product $y\, e^y = p \log p$ is the phase-space area of that prime in the window — lag times place. Neither factor alone collapsed the three L-functions: p gave the old $7p$ (local), $\log p$ is too slow, $(\log p)/\sqrt{p}$ is the bare tower weight and goes the wrong way. Stirling's $p \log p - p$ is the same area: $\Theta_v(\log p) \sim 1/p!$ would produce exactly this leading term. That reading is a mnemonic, not a derivation. What is structural is that the form multiplies an additive pairing by an Euler place, and the invariant of that product is w.

Why a Gaussian in w. Silence is linear in w: one cumulant, intensive in the speed s. Coupling is the remainder of $T_p v$ orthogonal to v. A quadratic cost in the same action is the next cumulant. Then $\kappa \sim e^{-c(\mu) w^2}$ is Gaussian in w, and c may depend on the window because the orthogonal of v grows with N. This is maximum-entropy bookkeeping, not a spectral theorem. It does explain, without a new parameter, why the margin is a parabola and why a single vector stops certifying around $p = 29$.

A check that would kill the reading: a fourth L-function whose $-ln\delta_p/(s\, w)$ is not 0.19, or a window at which $-ln\kappa$ is linear in w rather than quadratic. χ_5 or $\mu = 38$ on ζ would do.

Status

Tried. The three routes — decay at infinity, contour shift, edge continuity — miss e^{-7p} for the same structural reason: the lag is interior and the smallness is a dip of the Beurling extremal at arithmetic points. The measured laws live in the phase-space weight $w = p \log p$; the Gaussian in w for κ is the second cumulant of that action. Neither law is proved. The hole is the same hole as the quantitative floor. No RH.